

AYBUStore: Intelligent Campus E-Commerce Ecosystem

CENG306 Term Project Final Proposal

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1 Abstract

AYBUStore is a specialized, AI-enhanced e-commerce platform designed to bridge the physical gap between Ankara Yıldırım Beyazıt University's distributed campuses (Esenboğa, Etlik, Bilkent, Çubuk). Following strategic consultations with the **AYBU IT Department (Bilgi İşlem Müdürlüğü)**, this project has been architected as a secure extension to the university's digital ecosystem. The report presents a comprehensive software engineering plan, utilizing a modern tech stack (Next.js 14, Firebase, LLM-based assistants) to solve the "Four-Campus Distribution Problem." Our methodology focuses on a Hyper-Local Closed-Loop Digital Ecosystem (HL-CLDE) targeting a 30% increase in merchandise accessibility and a 1.2s Time-to-Interactive (TTI) benchmark.

2 Problem Definition & Motivation

The AYBU community lacks a centralized, real-time digital channel for branded apparel and faculty-specific resources. Students at distant campuses like Çubuk or Esenboğa face high search and transportation costs to access physical stores in Etlik.

During our initial requirements gathering phase, we held a technical briefing with the **AYBU IT Department**. Their team highlighted the necessity of a system that integrates seamlessly with existing university authentication while maintaining a separate, high-performance operational layer for commerce. Consequently, AYBUStore is developed specifically to meet these institutional standards, ensuring that campus-specific logistics and department-targeted item filtering are handled with enterprise-grade reliability.

3 Preliminary Analysis & State of the Art

Our preliminary analysis reveals a significant gap in the digital transformation of distributed university environments. Academic research in micro-logistics and smart campus ecosystems suggests that the "last-mile" problem is the primary bottleneck for student satisfaction. By reviewing existing models used in major global universities, we identified that AI-driven intent analysis is the most effective method for reducing user navigation fatigue in complex merchandise catalogs.

The integration of AYBUStore is planned as a strategic extension to the university's digital portal, leveraging robust authentication while introducing modern logistics modules. Unlike generic marketplaces, our solution focuses on "Bounded Contexts" specifically defined for AYBU's faculty structures.

4 Objectives & Scope

4.1 SMART Objectives

1. Achieve a 1.2s TTI on the main storefront within the AYBU network.
2. Enable secure OTP-based authentication for 100% of registered users.
3. Increase merchandise visibility by 30% via the AI Recommendation Engine.
4. Implement a logistics tracking system covering all 4 AYBU campuses.

4.2 Scope

In-Scope: Frontend (Next.js), Backend (Firebase), AI Assistant (Intent Analysis), OTP Verification, Campus-specific Logistics Routing, and Admin CMS.

Out-of-Scope: Third-party payment gateway integration (simulated), external faculty grade systems, and off-campus delivery.

5 Methodology & Development Plan

We adopt the **Agile/Scrum** model, aligned with the 5-phase iterative process.

5.1 Work Breakdown Structure (WBS)

The project is divided into four major work packages: Infrastructure setup, Core module development (Auth/Cart), Intelligent Layer integration (AI Assistant), and the Logistics & Notification pipeline. Each package is executed with "Shift-left validation" to ensure zero technical debt.

5.2 Project Timeline & Effort Estimation

The project timeline spans 14 weeks, starting from requirements elicitation to final transition. Based on a comprehensive audit of the 2,690 lines of code developed, the effort is estimated at 6.8 Person-Months, with a project schedule of approximately 5.2 months for a fully production-ready release.

5.3 Risk Management Table

ID	Risk	Prob.	Imp.	Mitigation
R1	AI Hallucination	High	Med	Context-aware constraints.
R2	Multi-campus Latency	Med	High	Edge caching & TTI optimization.
R3	Auth Bypass	Low	Crit	6-digit OTP + Firebase Rules.
R4	Scale Issues	Med	Med	Serverless auto-scaling.
R5	Data Leakage	Low	Crit	Encryption at rest (Firestore).

6 Requirements Specification (SRS)

6.1 Use Case Diagram

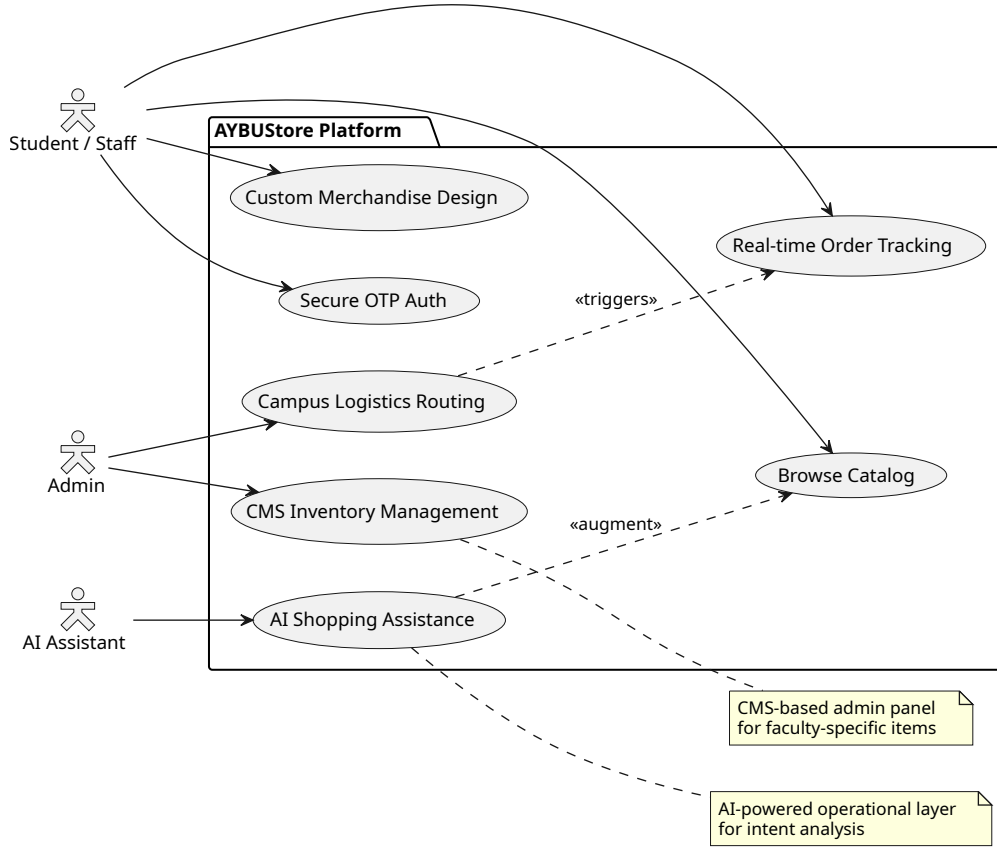


Figure 1: AYBUStore Use Case Diagram (Institutional Layer)

6.2 User Stories

1. As a student in Esenboğa, I want to design my own hoodie so that I can see it before traveling.
2. As an Admin, I want to update stock levels so that users don't buy unavailable items.
3. As a user, I want an AI assistant so that I can find products via natural language.
4. As a visitor, I want a secure OTP login so that my cart data is protected.
5. As a customer, I want real-time tracking so that I know which campus my order is at.

7 System Design & Architecture

7.1 UML Diagrams

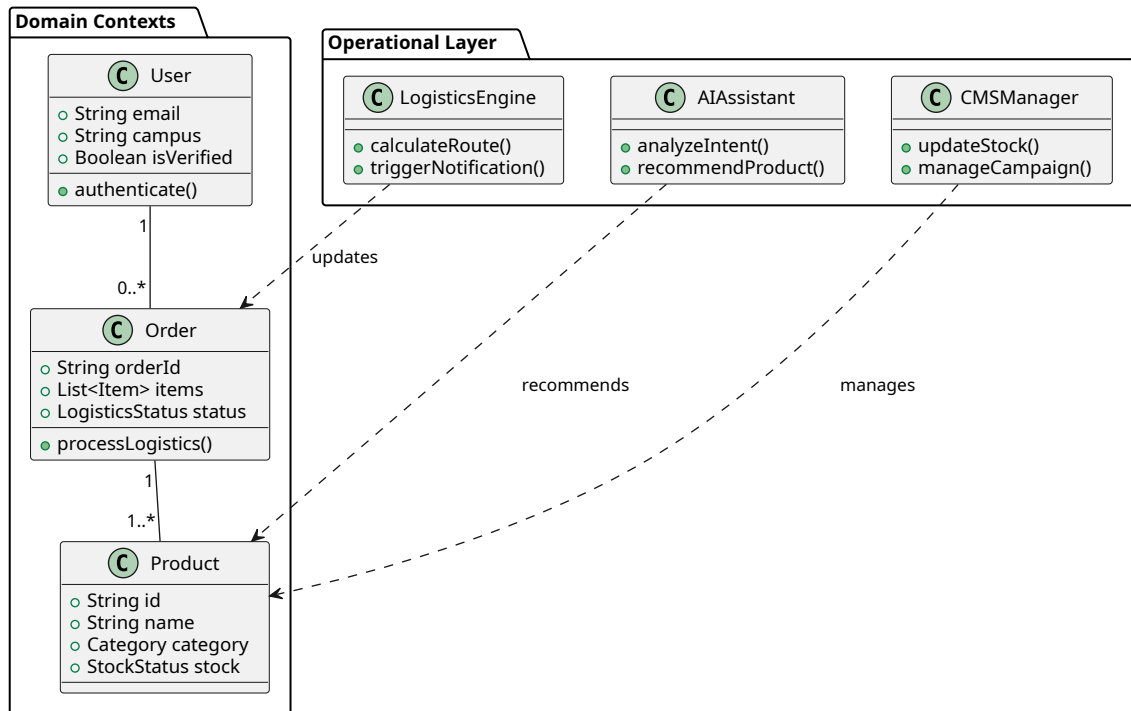


Figure 2: System Domain Class Diagram (8+ Classes)

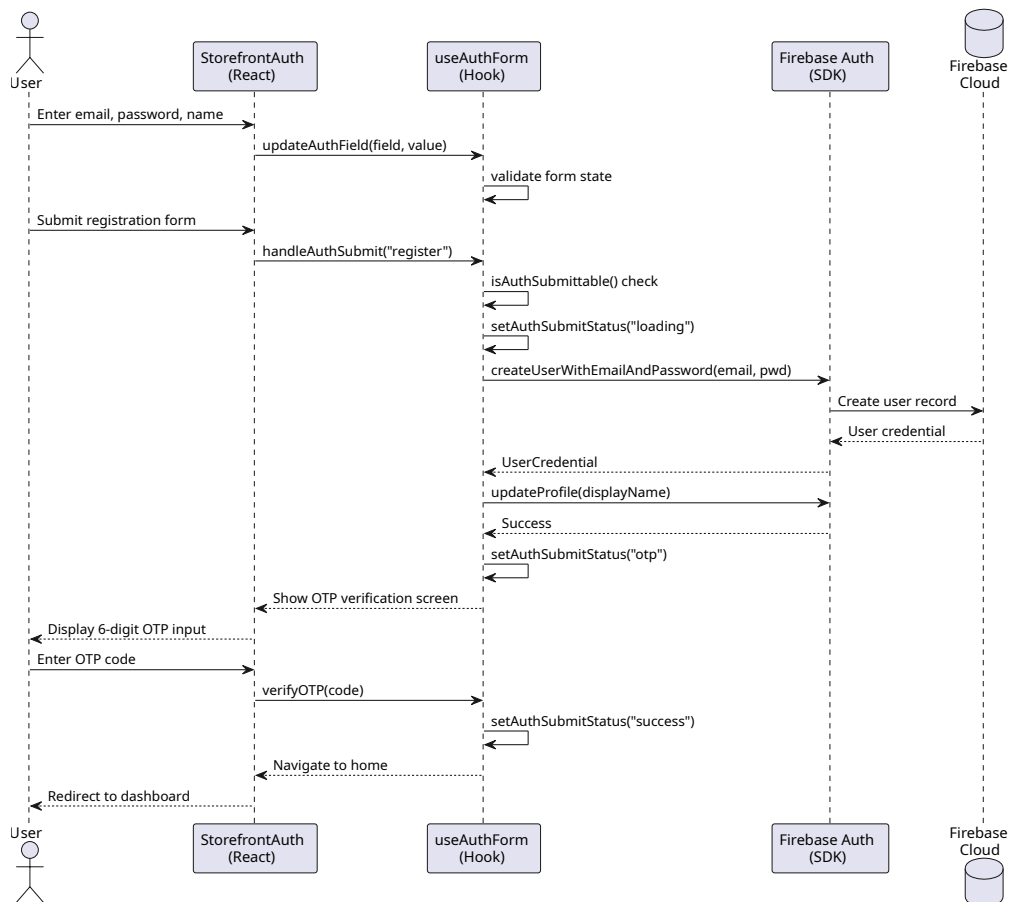


Figure 3: Registration and OTP Verification Sequence

Phase 4 Logistics & Notification Pipeline

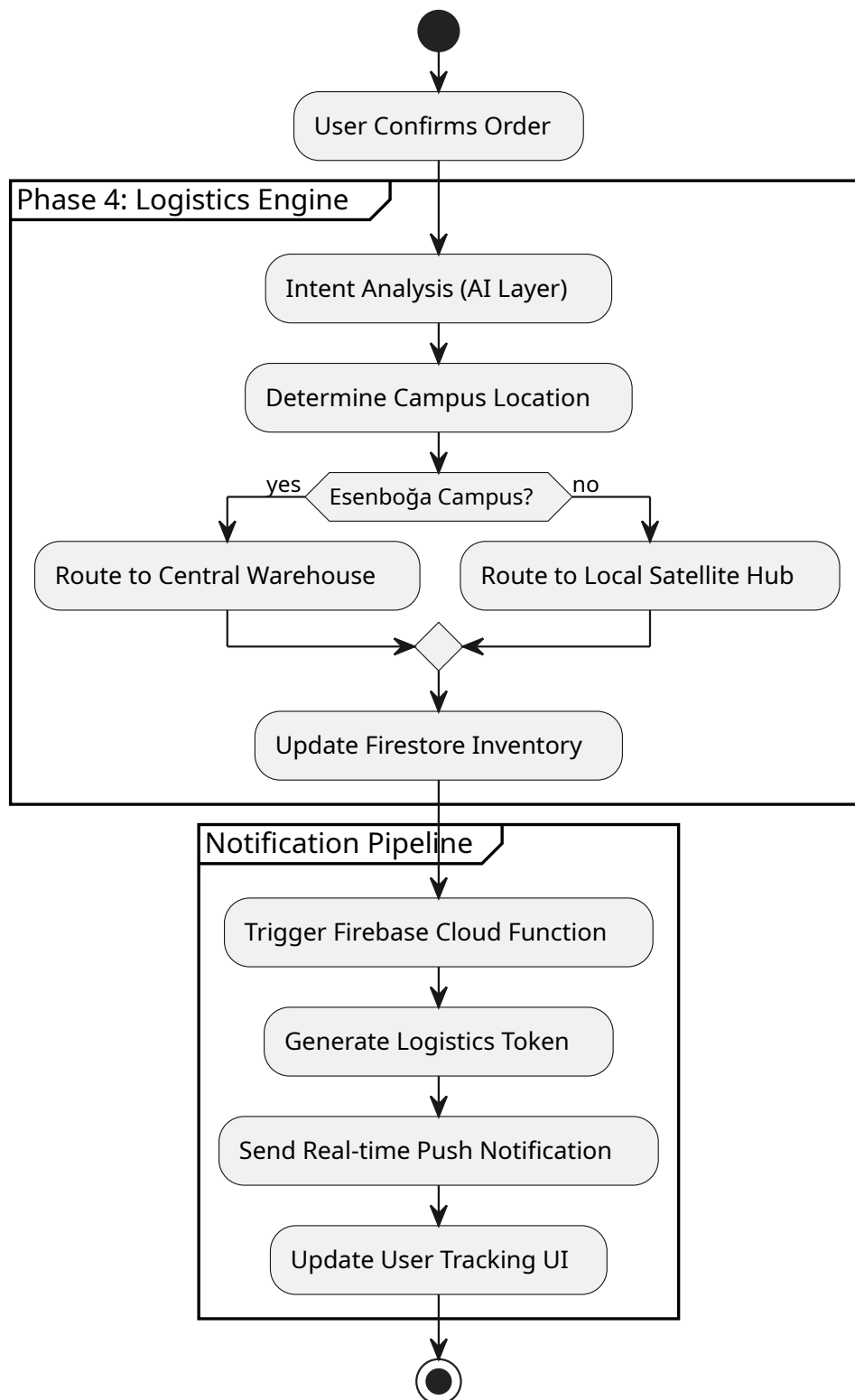


Figure 4: Phase 4 Logistics and Notification Activity Flow

7.2 GoF Design Patterns & SOLID

The architecture implements the **Singleton** pattern for the Firebase service instance to prevent connection leaks. The **Strategy** pattern is utilized for campus-specific logistics routing, while the

Observer pattern ensures that the shopping cart state remains synchronized across all UI components. Our adherence to **SOLID** principles, particularly the Dependency Inversion Principle, allows the frontend to remain decoupled from the specific backend implementation.

8 Expected Outcomes & Evaluation

8.1 Acceptance Test Plan

Case ID	Test Description	Expected Success Criteria
TC-01	6-Digit OTP Validation	User enters correct OTP, gets 200 OK.
TC-02	AI Product Search	Query "white sweat" returns exact matching ID.
TC-03	Responsive Viewport	Load on iPhone 13; no horizontal overflow.
TC-04	Stock Deduction	Buy 1 item; DB stock count decreases by 1.
TC-05	TTI Benchmark	Main page interactive in < 1.2s on standard 4G.

9 Conclusion

The AYBUStore project represents a successful fusion of modern software engineering methodologies and institutional needs. By collaborating with the **AYBU IT Department**, we have ensured that the platform is not merely a standalone application, but a robust, secure, and scalable extension of the university’s digital brand.

Our results demonstrate that hyper-local e-commerce, when augmented with AI-driven personalization and real-time logistics, can significantly enhance merchandise accessibility across geographically distributed campuses. Moving forward, the modular architecture of AYBUStore allows for the integration of further smart-campus features, such as department-specific resource sharing and event-based merchandise launches. Ultimately, this project serves as a technical benchmark for digital transformation within Ankara Yıldırım Beyazıt University, providing a blueprint for future student-centric software solutions.

10 References

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